

```

        fact=(n-j+beta)*term;
        numer[n-j]=numer[n-j+1]-fact*numer[n-j];
        denom[n-j]=denom[n-j+1]-fact*denom[n-j];
        term=term*ratio;
    }
    n++;
    val = abs(denom[0]) < small ? lastval : numer[0]/denom[0];
    lasteps = abs(val-lastval);
    if (lasteps <= eps) ncv++;
    if (ncv >= 2) cnvkd = 1;
    return (lastval = val);
}
};

```

You can use, or not use, the `cnvkd` flag exactly as previously discussed for `Epsalg`.

An alternative to the model sequence method of deriving sequence transformations is to use extrapolation of a polynomial or rational function approximation to a series, e.g., as in Wynn's ρ algorithm [1]. Since none of these methods generally beats the two we have given, we won't say any more about them.

5.3.3 Practical Hints and an Example

There is no general theoretical understanding of nonlinear sequence transformations. Accordingly, most of the practical advice is based on numerical experiments [5]. You might have thought that summing a wildly divergent series is the hardest problem for a sequence transformation. However, the difficulty of a problem depends more on whether the terms are all of the same sign or whether the signs alternate, rather than whether the sequence actually converges or not. In particular, logarithmically convergent series with terms all of the same sign are generally the most difficult to sum. Even the best acceleration methods are corrupted by rounding errors when accelerating logarithmic convergence. You should always use double precision and be prepared for some loss of significant digits. Typically one observes convergence up to some optimum number of terms, and then a loss of significant digits if one tries to go further. Moreover, there is no single algorithm that can accelerate every logarithmically convergent sequence. Nevertheless, there are some good rules of thumb.

First, note that among divergent series it is useful to separate out asymptotic series, where the terms first decrease before increasing, as a separate class from other divergent series, e.g., power series outside their radius of convergence. For alternating series, whether convergent, asymptotic, or divergent power series, Levin's u transformation is almost always the best choice. For monotonic linearly convergent or monotonic divergent power series, the ϵ algorithm typically is the first choice, but the u transformation often does a reasonable job. For logarithmic convergence, the u transformation is clearly the best. (The ϵ algorithm fails completely.) For series with irregular signs or other nonstandard features, typically the ϵ algorithm is relatively robust, often succeeding where other algorithms fail. Finally, for monotonic asymptotic series, such as (6.3.11) for $Ei(x)$, there is nothing better than direct summation without acceleration.

The v and t transformations are almost as good as the u transformation, except that the t transformation typically fails for logarithmic convergence.

If you have only a few numerical terms of some sequence and no theoretical insight, blindly applying a convergence accelerator can be dangerous. The algorithm